

QUESTION #01

i) the transport layer provides logical, not physical, communication between application processes because it presents each application with a virtual endpoint identified by an IP address + a port number. It performs segmentation and reassembly so applications see a continuous byte stream, and it multiplexes/demultiplexes flows using (host IP, port) pairs so applications are insulated from link types, physical cables, and hops. (ILLUSTRATION AT END)

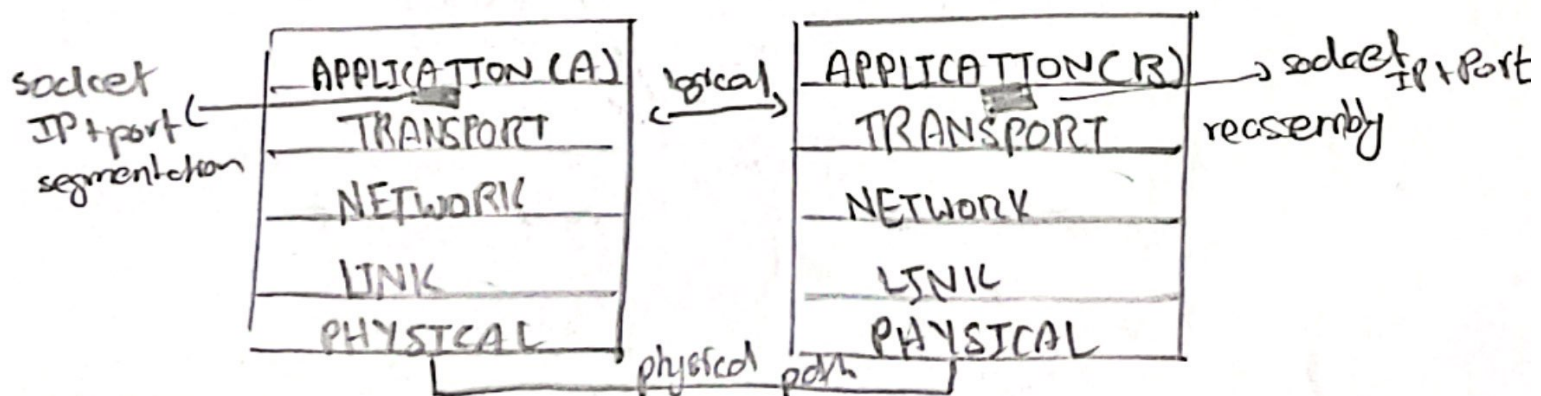
ii) UDP is preferred when low latency and low overhead matter, retransmissions are undesirable, or multicast is required, for example, DNS queries, VoIP and real-time audio/video (RTP), live streaming, online multiplayer games, DHCP, and IPTV.

iii) rdt-revc() is the receiver-side routine called when a packet arrives from the unreliable lower layer. It checks checksum and sequence number, delivers correct, new data to the application, updates expected sequence, and sends an Ack back. rdt-send() is the sender-side routine invoked by the application to make a packet, attach sequence number and checksum, store it in the sender buffer, call udt-send() to push it to the network, and start retransmission timers, so rdt-rcv() handles incoming packet processing and Ack generation, while rdt-send() handles outgoing-packet creation and transmission.

iv) udt-send() is the abstraction of the unreliable network used by RDT implementations. It takes a packet from the sender and injects it into the simulated/unreliable lower layer where the packet may be lost, corrupted, duplicated, or delayed, therefore it isolates the protocol logic from the network's misbehavior.

v) Software-Defined Networking separates the control plane from the dataplane, allowing centralized, programmable network management through an SDN controller. It enables automation, flexibility, and efficient traffic control using protocols like OpenFlow.

vi) Traditional networks have distributed control and manual configuration, while SDNs have centralized control and software-based management. SDNs are programmable, flexible, and easier to scale compared to rigid, hardware-driven traditional networks.



QUESTION #02

max datagram = 1500 bytes (including header)

IP header = 20 bytes

MP3 file size = 48840 bytes

Payload per datagram = $1500 - 20 = 1480$ bytes

Number of datagrams = $\frac{48840}{1480} \approx 33$ exact RT

QUESTION #03

2314 - 0914 $\rightarrow$ X = 0, Y = 9, Z = 1

substituting $\rightarrow$ $\overset{Y}{0009} \overset{Z}{0010} 0000 0001 0000 0000 3FF0 0000 \overset{X}{0000}$
 $FFFF 0100 0000 0010 0000$

- ① Source Port $\rightarrow 0009 = 9$
- ② Destination Port $\rightarrow 0010 = 16$
- ③ Sequence Number $\rightarrow 00000001 = 1$
- ④ Acknowledgement Number $\rightarrow 00000000 = 0$
- ⑤ TCP Header Length $\rightarrow 3FF0$ (data offset is first 4 bits = 3) = $3 \times 4 = 12$ bytes
 $0011 1111 1111 0000$
- ⑥ ACK Bit $\rightarrow 1$
- ⑦ FIN Bit $\rightarrow 0$
- ⑧ SYN Bit $\rightarrow 0$
- ⑨ Window Size $\rightarrow 0000 = 0$
- ⑩ Checksum $\rightarrow 0000 = 0$